



SpiderRock V7 Tool Concept Guide

Symbol Viewer

November 1, 2019

SYMBOL VIEWER

The SpiderRock Symbol Viewer is a live order entry and risk management tool at the symbol level, allowing users to trade simple DMA style as well as complex equities, options and futures orders. The tool provides access to all SpiderRock algorithmic execution logic and displays live prices and analytics for all products supported by the platform. It also displays real time position risk records and has graphical capabilities to display option analytics as well as position greeks. This document is a concept guide, and provides an overview of the tools main features. For more details, please contact the support desk at cs@spiderrock.net or reference other related concept guides.



 LAUNCH SAVED LAYOUT

 SAVE CURRENT LAYOUT AS DEFAULT

SYMBOL VIEWER

- View live market data and position-risk for equities, options, and futures
- Enter and manage SpiderRock simple and complex parent orders and baskets
- Access SpiderRock live surface analytics and charts

SYSTEM INFORMATION

User: c.sampat.2 (ClientTrader)
Version: v7.0
Citrix Host: SPIDERNJ131
Tool Server: 10.37.201.161:5240
Environment: V7_Stable.Prod | NY4
Time: 10:16:14.1 AM



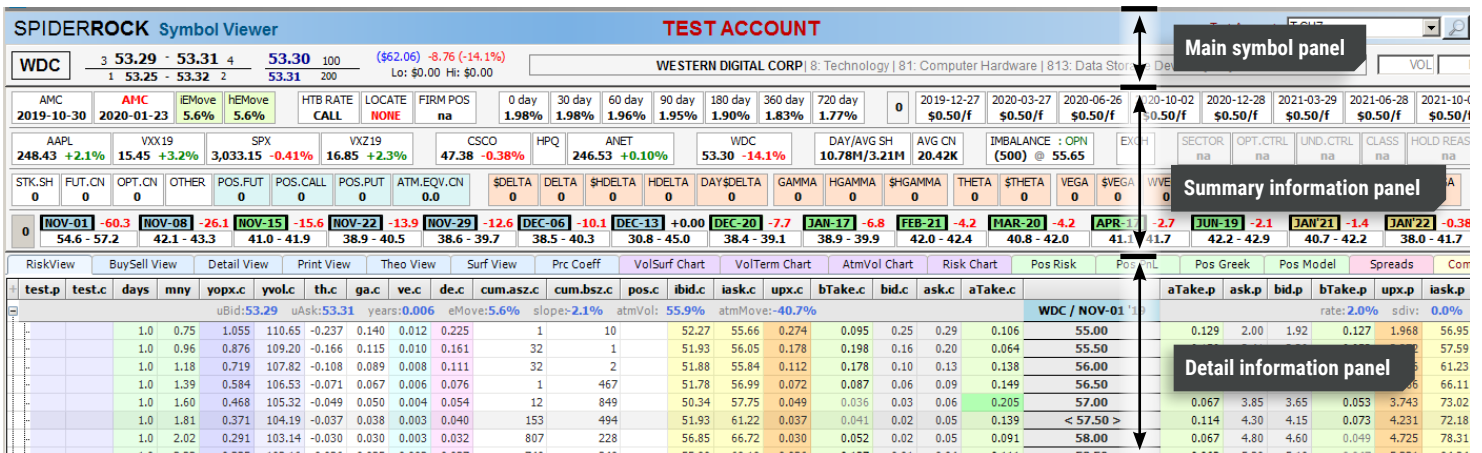
The table below is a description summary of each tab available via the Symbol Viewer. Only the actionable tabs (RiskView and Spreads) and chart tabs are described in this document. For a complete store list of available metrics, please contact the support desk at cs@spiderrock.net.

VIEW/TAB	STORE	DESCRIPTION
RiskView	Option Pair	Option montage, which includes real-time market data and analytics, dividend streams, earnings, interest rate, and more. User can launch order entry tickets from this window (quick ticket or ticket blotter) for all instruments supported
BuySell View	Option Pair	Provides theoretical edge data to aid in trading decisions. tEdg.c/p lights up in green/red when there is an indication of a potential trade (when theo b/s price mid-point is through the market bid/offer)
Detail View	Option Pair	Provides contract deliverable information, bid/ask sizes with corresponding exchanges, and an indication of total option volume traded and time at which quote last updated
Print View	Option Pair	Provides open interest, print and volume information by series as quote width for a particular series
Theo View	Option Pair	Provides data on various theoretical pricing measures. This tab is for Users who load their own theoretical values
Surf View	Option Pair	Displays theoretical volatility surface data.
Prc Coeff	Option Pair	Displays theoretical pricing data.
VolSurf Chart	Charts	Displays current Implied Volatility surface
VolTerm Chart	Charts	Displays term structure of ATM Implied Volatility
AtmVol Chart	Charts	Displays 5-minute graph of ATM Implied Volatility
Risk Chart	Charts	Displays PnL, Delta, Gamma, Theta, and Vega
Pos Risk	Risk	Provides risk slides and estimated event P&L effects by series
Pos PNL	Risk	Breaks down various measures of P&L by series and provides mark data and P&L allocation for today's trades in a particular position
Pos Greek	Risk	Current and previous day Greek calculations
Pos Model	Risk	Theoretical pricing model inputs
Spreads	Option Pair	Actionable spread filtering window for a particular instrument
Comments	N/A	Freeform text

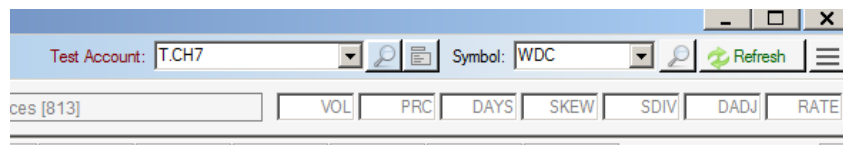


RISKVIEW TAB

- The RiskView tab is an option montage screen, which provides pricing information (NBBO prices, implied vols, alpha probabilities), dividend streams, earnings, interest rates, hard to borrow rates, greeks and many other metrics users can select from right clicking on any column header below the tab selectors.
- The window has three distinct parts: the main symbol panel, the summary information panel, and the detail information panel as shown below.



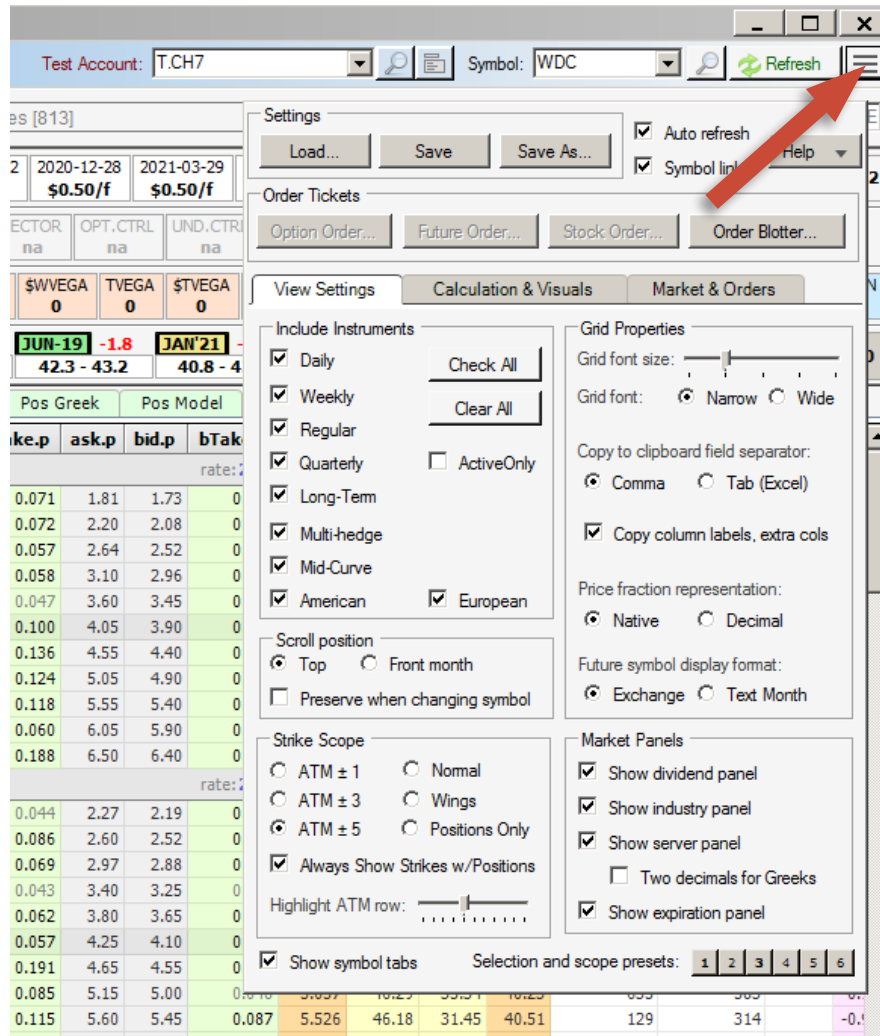
- The main symbol panel (upper right corner) allows users to select one or several SpiderRock accounts, an asset/symbol, and "what if" overrides to drive the analytics in the detail window of the tool. The Refresh button loads the data.



TAG	DESCRIPTION
VOL	User can enter a flat volatility override that will apply to all strikes and expirations in the Risk View. Both 'user' columns as well as Greek columns will use the override for calculation purposes
PRC	User can enter a specific underlying price from which Risk View will use to calculate greeks and P&L
DAYS	A days ADJUSTMENT, added or removed from the days to expiration. For example, if there were 5 days to expiration and you entered 1 then it would show the effect of 6 days to expiration (5+1) and likewise if you entered -1 then it would show the effect of 4 days to expiration (5-1)
SKEW	Adjust the skew of the volatility curve
SDIV	This value is 0 for general collateral (ETB) names. As it gets harder to borrow, the SPIDERROCK system calculates a percentage measure of how much harder it is to borrow. User can adjust this across all options and is reflected in the sdv column of the [Risk View] tab.
DADJ	This allows User to enter a multiple for the dividend stream (DDIV). All dividends in the stream will be multiplied by the amount entered
RATE	Allows User to override prevailing Interest Rate for the underlying displayed

RISKVIEW TAB

- Clicking on the setting button as shown below by the red arrow, brings a menu where users can set a variety of view settings, pricing calculations and heat map preferences, as well as quick order ticket parameters and Bloomberg Links. For more information on these settings, please contact the support desk at cs@spiderrock.net.



RISKVIEW TAB

- The upper left corner of the main symbol panel displays pricing and size information for the underlying symbol as shown below. Double clicking on the NBBO or size will launch a stock or a futures order entry ticket.

SPIDERROCK Symbol Viewer									
AAPL	5	170.24	-	170.25	8	170.24	500	(\$171.10)	-0.86 (-0.50%)
	13	170.23	-	170.26	2	170.25	400	Lo: \$169.83	Hi: \$171.39
AMC	AMC	iEMove	hEMove	HTB RATE	LOCATE	FIRM POS	0 d		
2017-11-02	2018-01-30	6.2%	4.4%	CALL	NONE	NA	1.1		

- The **summary information panel** displays the dividend stream, industry related information such as last price and % move for similar symbols, and other server information such as position greeks and PnL metrics.

HTB RATES / DIVIDEND PANEL

HTB RATES							DIVIDENDS									
0 day	30 day	60 day	90 day	180 day	360 day	720 day	0	2017-11-10	2018-02-08	2018-05-07	2018-08-09	2018-11-08	2019-02-09	2019-05-08	2019-08-10	2019-11-07
1.18%	1.24%	1.27%	1.32%	1.46%	1.73%	1.82%		\$0.63/e	\$0.63/f	\$0.65/f	\$0.65/f	\$0.65/f	\$0.65/f	\$0.69/f	\$0.69/f	\$0.69/f

TAG	DESCRIPTION
AMC	Prior earnings date
AMC / BMO	Next earnings date - in this case BMO (before market open)
iEMove / hEMove	Implied Earning Move and Historical Earning Move. By clicking on these boxes, one can open up a new dialog window to show more historical information for each value.
HTB RATE	Hard to borrow rate. In this example, it is "GC" means "General Collateral which effectively means it is easy to borrow. If there was a rate in this field, this will be the fee charged on a pro-rate basis depending on how many days you hold the position short. It will be assessed to your account at the end of the month upon settlement of the closing trade.
LOCATE	The clearing firms current inventory
FIRM POS	Stock position as an aggregate for the firm as well as in the account displayed
0-720 DAYS	The potential HTB rates out of 720 trading days
DIVIDENDS	The Dividend dates and amounts. If not dividends, then the legend "NO DIVIDENDS" will be shown



RISKVIEW TAB

INDUSTRY PANEL

SELECTED SYMBOLS				INDUSTRY SYMBOLS			ACTIVE SYMBOL
SPY 247.05 +0.08%	IWM 139.11 -0.22%	SPX 2,468.62 +0.08%	VIX 11.55 -1.0%	NWL 44.86 -4.9%	HAS 92.50 -1.2%	WHR 168.96 -0.73%	NFLX 178.97 -0.16%

TAG	DESCRIPTION
Selected Symbols	Users are able to customise their watchlist in this panel. You can select up to 4 symbols by right-clicking on any of these symbol boxes and then select "Select Symbols" menu option and enter in the 4 symbols you always want to show. These will always be shown irrespective of the primary symbol selected.
Industry Symbols	Up to 3 industry-related symbols. In this example, our primary symbol is F (Ford) so therefore it is also showing me TSLA, GM, and JCI. The system always selects these symbols.
Active Symbols	Information regarding the existing primary (selected) symbol.



By right clicking on any of the selected symbols, the user is able to open a quick ticket and send trades in that symbol.

SERVER PANEL

SHARES / CONTRACTS							GREEKS							
STK.SH 0	FUT.CN 0	OPT.CN 378	OTHER	POS.FUT 0	POS.CALL 378	POS.PUT 0	\$DELTA 4,729,198	DELTA 29,331	\$HDELTA 4,731,822	HDELTA 29,345	GAMMA 611	THETA -2,393	VEGA 7,760	WVEGA 2,050

PnL					RISK SLIDES						
SIM PNL na	EDGE \$0	EDG/CN \$0.00	WDTH \$3,581	DAY PN \$0	VOL PNL \$-23,019	DN 50% \$-573,819	DN 15% \$-459,333	DN 8% \$-305,190	UP 6% \$318,591	UP 15% \$896,922	UP 50% \$3,859,040

Users can customize all above values. These values are an example from possible choices.

- The detail information panel is highly customizable by the user. Screen colors and metrics are all configurable by right clicking on any column header right below the tab selectors.
- Users can launch a simple order entry ticket (as shown on the next page) for stocks, futures and options by double clicking on any bid/ask prices.



RISKVIEW TAB

- Users can also launch complex orders in stocks, futures and options by right-clicking on any bid/ask price, or highlighting several strikes and right-clicking. This action will display the following menu:

RACE / NOV-01 '19					
17	10.50	14.10	0.036	148.00	0.030
73	9.90	12.50	0.027	149.00	0.040
87	9.40	11.30	0.086	150.00	0.031
31	6.70	8.40	0.033	152.50	0.024
33	4.50	5.30	0.034	155.00	0.026
56	2.45	2.45			
98	0.70	0.70			
42	0.05	0.05			
3id	0.00	0.00			
3id	0.00	0.00			
3id	0.00	0.00			
19	12.70	12.70			
80	11.80	11.80			
91	10.90	10.90			
83	8.90	8.90			
30	7.00	7.00			
83	5.30	5.30			
80	3.90	3.90			
27	2.80	2.80			
75	1.90	1.90			
94	1.20	1.20			
44	0.75	0.75			

Spider Option Ticket (New)
☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6
Spider Option Ticket (Add)
☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6
Spider Option MLeg Ticket (New)
☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6
Spider Option MLeg Ticket (Add)
☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6
Spider Option Ticket / Close (Add)
Spider Option Ticket / Buy.Close (Add)
Spider Option Ticket / Sell.Close (Add)
NBBO Quotes (New)
NBBO Quotes (Add)
Underlying Spider Ticket (New)
Underlying Spider Ticket (Add)

- » User can choose to create a new option or multi-leg ticket or add to an existing option or multi-leg ticket
- » Radio buttons (1, 2, 3, 4, 5, 6) are available in this menu to recall pre-saved settings for option and multi-leg tickets
- » User can also select a closing strategy, launch a ticket on NBBO quotes, or launch a ticket in a cooresponding underlyer.

- Users can view positions in a particular symbol from that screen and corresponding greeks. Users can also simulate positions in the test columns and visualized the corresponding simulated greeks as they turn pink from the initial orange color.

\$DELTA	DELTA	\$HDELTA	HDELTA	GAMMA	THETA	VEGA	WVEGA	SIM PNL	EDGE	EDG/CN	WDTH	DAY PN
6,235,282	38,671	6,238,053	38,684	773	-3,351	8,075	2,153	na	\$0	\$0.00	\$3,581	\$0

11

DEC-15

-0.18

JAN-19

-0.21

FEB-16

-0.08

APR-20

-0.07

JUN-15

-0.06

23.8 - 23.9

23.4 - 23.5

24.4 - 24.4

24.0 - 24.1

24.5 - 24.5

View	Surf View	Prc Coeff	VolSurf Chart	VolTerm Chart	AtmVol Chart	Risk Chart	Pos Risk	Pos PnL				
px.c	cum.bs.c	bTake.c	ibid.c	bid.c	ask.c	ias.c	aTake.c	cum.as.c	pos.c	test.c		test.p
uAsk: 161.2500 rate: 0.012 atmVol: 0.161											AAPL / SEP-08 '17	
6.283	1,406	0.073		6.15	6.35	41.5	0.099	289	+10		155.00	
0.026	64	0.629	21.4	0.03	0.04	22.6	0.165	2,104	+2		< 165.00 >	
uAsk: 161.2500 rate: 0.012 atmVol: 0.260											AAPL / SEP-15 '17	
2.536	1,044	0.060		12.20	12.60	35.5	0.109	673			149.00	
1.576	17	0.131	24.8	11.40	11.60	33.1	0.195	1		+100	150.00	
9.229	10	0.192	27.2	9.10	9.25	30.8	0.121	341			152.50	
7.014	636	0.223	26.8	6.90	7.00	28.5	0.266	477			155.00	
5.011	30	0.555	26.7	4.95	5.00	27.3	0.320	5			157.50	

RISKVIEW TAB

- Launching order entry tickets:

Quick tickets A single order can be launched from left-double clicking on any bid/ask prices (or implied volatilities) from the options montage screen, or any bid/ask prices for the underlying instrument in the upper right corner of the screen. A number of settings are available to users to launch an order, including order size, price or volatility limit, order type, limit class, max visible, number of exchanges (when posting liquidity), cancel overlap, auto hedge strategy, time-in-force, and more.

SPIDERROCK Option Order Test Account: TCH7 | TEST

RACE 1 159.75 - 159.83 5 159.75 100 (\$158.14) +1.61 (+1.0%)
 1 159.74 - 159.84 1 0.00 0 Lo: \$159.16 Hi: \$160.33

FERRARI N V 0: unknown | 0: unknown | 0: unknown [0]

call market (ISE) 199 4.50 - 5.20 73 (MIAX) | 3.40/ 1 (CBOE, 1min) | - - 42.52
put market - - - 0.15 279 (MIAX) | 0.15/ 5 (MPRL, 1min) | - - 30.85

RACE / NOV-01 2019 / \$155 C uPrc: \$159.750, de: +0.941 valid range: \$151.76 - \$167.74 1.0% 2.0% 5.0% sdv: 0.00%, ddiv: \$0.00 Exchanges (All)

Quantity / Prc Limit Order Type / Post Type Limit Class Max Visible
 5.20 @ 42.52% PostOnly PostWith Simple Prc 5,000 NumExch: 1 CxlOverlap BUY Order Status / Open / Filled NONE 0 0 Switch to SELL

Auto Hedge Time In Force Take Reach / Size Release Probability Limit Surface Limit
 NONE 4:00 pm Immediate Immediate Static MakeLimit: 0.00 TakeLimit: 0.00 PrcOffset: 0.00% VolOffset: 0.00% Opx Floor: 0.00 Public Size: Default

Ticket blotter Order baskets can be launched from highlighting any strikes on the options montage screen across multiple expiries and multiple symbols if desired, then right clicking and choosing Spider Option Ticket (New). This will populate the ticket blotter with all selected series. All settings from the quick ticket and more are available to users to launch an order basket. In addition, users can apply risk groups that act as guardrails when executing the basket. These risk groups can be expressed as vega limits, premium limits, contract limits and/or dollar delta limits.

SPIDERROCK Ticket Blotter Test Account: TCH7 | TEST

Stock Future Option MultiLeg

Order Type / Post Type Limit Class Probability Limit Surface Limit Exchanges
 ActiveTaker PostLimit Simple Prc Static MakeLimit: 0.50 TakeLimit: 0.50 PrcOffset: 0.000 VolOffset: 0.00 % Exchanges (All) Send New Update Existing

Auto Hedge Time In Force Limit Prc Adjuster (value / reference) Take Reach / Size Release Max Visible
 NONE 4:00 pm 0.00 abs / surf +bid +ask Immediate Immediate CxlOverlap 100 % all NumExch: 1 PublicSize: Default Opx Floor: 0.00 UPrange (%): -2.00 2.00

	RACE 08-NOV-2019	Side	Size	uLimitPrc	Open	Filled	State	Limit	oBid	oAsk	iBid	iAsk	uBid	uAsk	sVol	sPrc	optVe	optDe	optGa	Pos	Bot	Slid
X	149.00 CALL	BUY	0	12.00	0	0	NONE		11.60	12.00	41.40	48.16	159.77	159.83	45.28	11.823	0.060	0.846	0.021	0	0	0
X	150.00 CALL	BUY	0	11.10	0	0	NONE		10.80	11.10	42.03	46.68	159.77	159.83	44.61	10.963	0.065	0.826	0.023	0	0	0
X	152.50 CALL	BUY	0	9.10	0	0	NONE		8.80	9.10	41.63	45.49	159.77	159.83	43.13	8.916	0.077	0.766	0.028	0	0	0
X	155.00 CALL	BUY	0	7.20	0	0	NONE		6.90	7.20	40.26	43.64	159.77	159.83	41.89	7.044	0.089	0.692	0.033	0	0	0
X	157.50 CALL	BUY	0	5.50	0	0	NONE		5.30	5.50	39.96	42.02	159.77	159.83	40.82	5.383	0.097	0.604	0.037	0	0	0
X	160.00 CALL	BUY	0	4.00	0	0	NONE		3.90	4.00	39.26	40.25	159.77	159.83	39.87	3.962	0.101	0.508	0.040	0	0	0
	RACE 15-NOV-2019	Side	Size	uLimitPrc	Open	Filled	State	Limit	oBid	oAsk	iBid	iAsk	uBid	uAsk	sVol	sPrc	optVe	optDe	optGa	Pos	Bot	Slid
X	152.50 CALL	SELL	0	9.10	0	0	NONE		9.10	9.40	33.65	36.46	159.77	159.83	34.84	9.226	0.107	0.753	0.027	0	0	0
X	155.00 CALL	SELL	0	7.30	0	0	NONE		7.30	7.60	33.18	35.67	159.77	159.83	33.96	7.393	0.120	0.682	0.031	0	0	0
X	157.50 CALL	SELL	0	5.70	0	0	NONE		5.70	5.90	32.73	34.26	159.77	159.83	33.27	5.771	0.130	0.600	0.034	0	0	0
X	160.00 CALL	SELL	0	4.20	0	0	NONE		4.20	4.50	31.41	33.64	159.77	159.83	32.71	4.376	0.135	0.512	0.036	0	0	0
X	162.50 CALL	SELL	0	3.10	0	0	NONE		3.10	3.30	31.37	32.88	159.77	159.83	32.21	3.211	0.132	0.421	0.036	0	0	0

Vega Limits (Buy | Sell | Abs) Premium Limits (Buy | Sell | Abs) Contract Limits (Buy | Sell | Abs) Dollar Delta Limits (Buy | Sell)
 L: none S: none At: none R: 1.00 L: none S: none At: none L: none S: none At: none L: 150,215 S: 150,215 Auto AutoSize MktSize

RACE / 2A58 <none> / D09D

For further information, please request the **Order Entry Tickets concept guide** or contact the support desk at cs@spiderrock.net.



SPREADS TAB

- The spreads tab allows users to apply filters to spread feeds so that corresponding spreads waterfall in the display window. Users can then select a particular spread from the display window and launch a spread order entry ticket to trade the spread.
- SpiderRock currently consumes the CBOE, ISE and PHLX spread feeds for equities, and the CME, ICE and CFE spread feeds for futures.
- Users can select a combination of different filters. The main filter buckets are surface, edge, quantity, probability, spread type, spread ratio, moneyness, time, nature of the spread components, source, and exchange.
- Once filters are selected, and corresponding spreads waterfall in the display window, users can double left-click on a line (see red arrows below), which populates an order entry ticket.
- The order entry ticket allows users to select order type, order size, order price, limit class, probability limit, auto-hedge, time-in-force, leg exposure and more. For more information on the spread ticket, please contact our customer support desk at cs@spiderrock.net.

SPIDERROCK Symbol Viewer **TEST ACCOUNT**

AAPL 1 256.61 256.62 4 256.61 116 (\$257.13) -0.52 (-0.20%) APPLE INC | 3: Consumer Goods | 31: Consumer Durables | 3

2019-10-30	2020-01-28	4.6%	4.6%	HTB RATE	LOCATE	FIRM POS	0 day	30 day	60 day	90 day	180 day	360 day	720 day	2019-11-07	2020-02-06	2020-05-07	2020-08-06	2020-11-05	2021-02-05	2021-05-07	2021-08-06
256.61	-0.20%	15.00	-0.17%	3,073.41	-0.04%	16.54	-0.21%	94.45	-1.4%	48.80	-3.5%	20.09	+0.58%	0	\$0.77/e	\$0.77/f	\$0.81/f	\$0.81/f	\$0.81/f	\$0.85/f	\$0.85/f

STK_SH 0 FUT_CN 0 OPT_CN 0 OTHER 0 POS_FUT 0 POS_CALL 0 POS_PUT 0 ATM_QC 0 SDELTA 0 DELTA 0 SHDELTA 0 HDDELTA 0 DAYDELTA 0 DAYSDDELTA 0 GAMMA 0 HGAMMA 0 SHGAMMA 0 DAYGAMMA 0 THETA 0 SHTHETA 0 DAYTHETA 0 VEGA 0 WVEGA 0

Filter window

Text Search: [] Active only []
 Min Expiry Date: [All dates]
 Surface Delta: 0.00 .. 1,000.00
 Surface Vega: 0.00 .. 1,000.00
 Spread: Any [x] Single [x] RevCnv [x] Straddle [x] Strangle [x]
 Ratio: Any [x] 1x1 [x] 1x2 [x] 1x3 [x]
 Time: Any [x] 1w [x] 1m [x] 3m [x] 6m [x] > 6m [x]
 Legs: Any [x] Stocks [x] Futures [x] Options [x]

Display window

BidMask	BidExch	AskExch	AskMask	SurfGamma	Description	OptLegs	Type	Ratio	MinExpiry	MaxYears	MinYears	UPrc	Group	SubIndustry	Industry	MktAskGap	MktBidGap	SurfEdge	TrickerSrc
I	ISE	ISE	I	-0.02	AAPL:15-JAN-2021 240,000 C	2	Vertical	R1x1	2019-11-15	0.03	0.03	256.63	314 Electro	31 Consum	3 Consume	0.10	-0.30	-0.06	NMS
B	BOX	PHLX	X	0.03	AAPL:15-JAN-2021 240,000 C	2	Strangle	R1x1	2020-01-17	0.03	0.03	256.63	314 Electro	31 Consum	3 Consume	14.70	-0.15	14.68	NMS
X	PHLX	PHLX	X	-0.03	AAPL:15-JAN-2021 240,000 C	2	Time	R1x1	2019-11-08	0.03	0.03	256.63	314 Electro	31 Consum	3 Consume	-1.80	-1.80	-1.90	NMS
X	PHLX	PHLX	X	0.00	AAPL:15-JAN-2021 240,000 C	3	OtherOpt	None	2019-11-15	0.00	0.00	256.63	314 Electro	31 Consum	3 Consume	-102.90	-103.23	-103.23	NMS
X	PHLX	PHLX	X	0.00	AAPL:15-JAN-2021 240,000 C	2	Vertical	R1x1	2021-01-15	1.19	1.19	256.69	314 Electro	31 Consum	3 Consume	-0.57	-0.06	-0.06	NMS

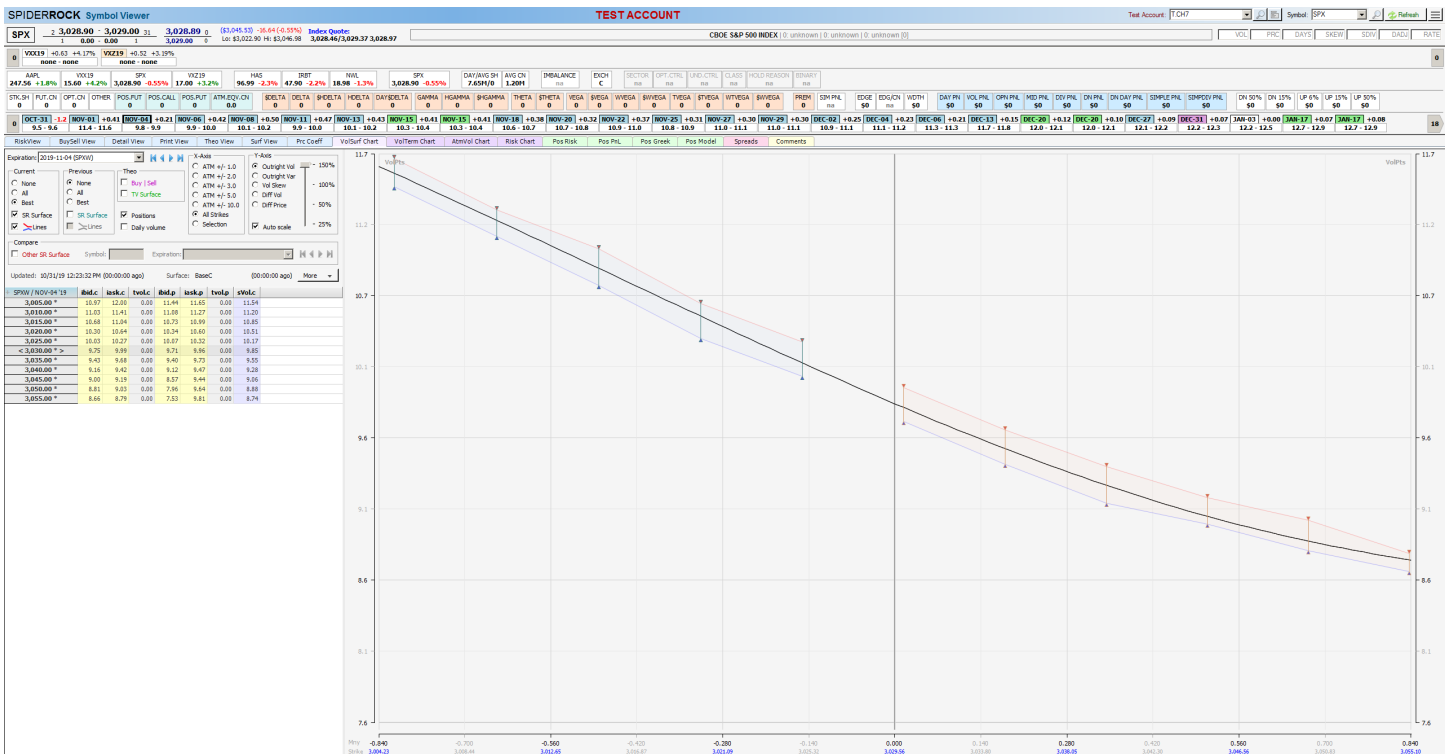
Spread ticket

Stock: AAPL 08-NOV-2019 220.00 C
 Order Type: Post Limit
 Order Size: 10
 Order Price: 0.000
 Limit Class: Simple
 Probability Limit: Static
 Auto Hedge: NONE
 Time In Force: 1 hour
 Stock Leg / Ref Price: 0.000
 Exposure / Slippage / Lead: 20% 0.00 Any
 Size Release: Immediate
 Max Visible: 100
 NumExch: 3

Option Market:	Market w/Stock:	Live Spread Market:	LegSide	LegRatio	uLimitPrc	effLmtPrc	Open	Filled	Limit	State	Child Orders	BidPrc	AskPrc	Bid
X	AAPL 08-NOV-2019 220.00 C	SELL	1	1.250	1.250	0	0	0	0	NONE		36.60	36.65	
X	AAPL 08-NOV-2019 222.50 C	SELL	1	(1.250)	(1.250)	0	0	0	0	NONE		34.05	34.20	
		BUY	0	0.000	0.000	0	0	0	0	NONE		256.61	256.62	
			0	0.000	0.000	0	0	0	0	NONE		(70.85)	(70.65)	
			0	0.000	0.000	0	0	0	0	NONE		?	?	

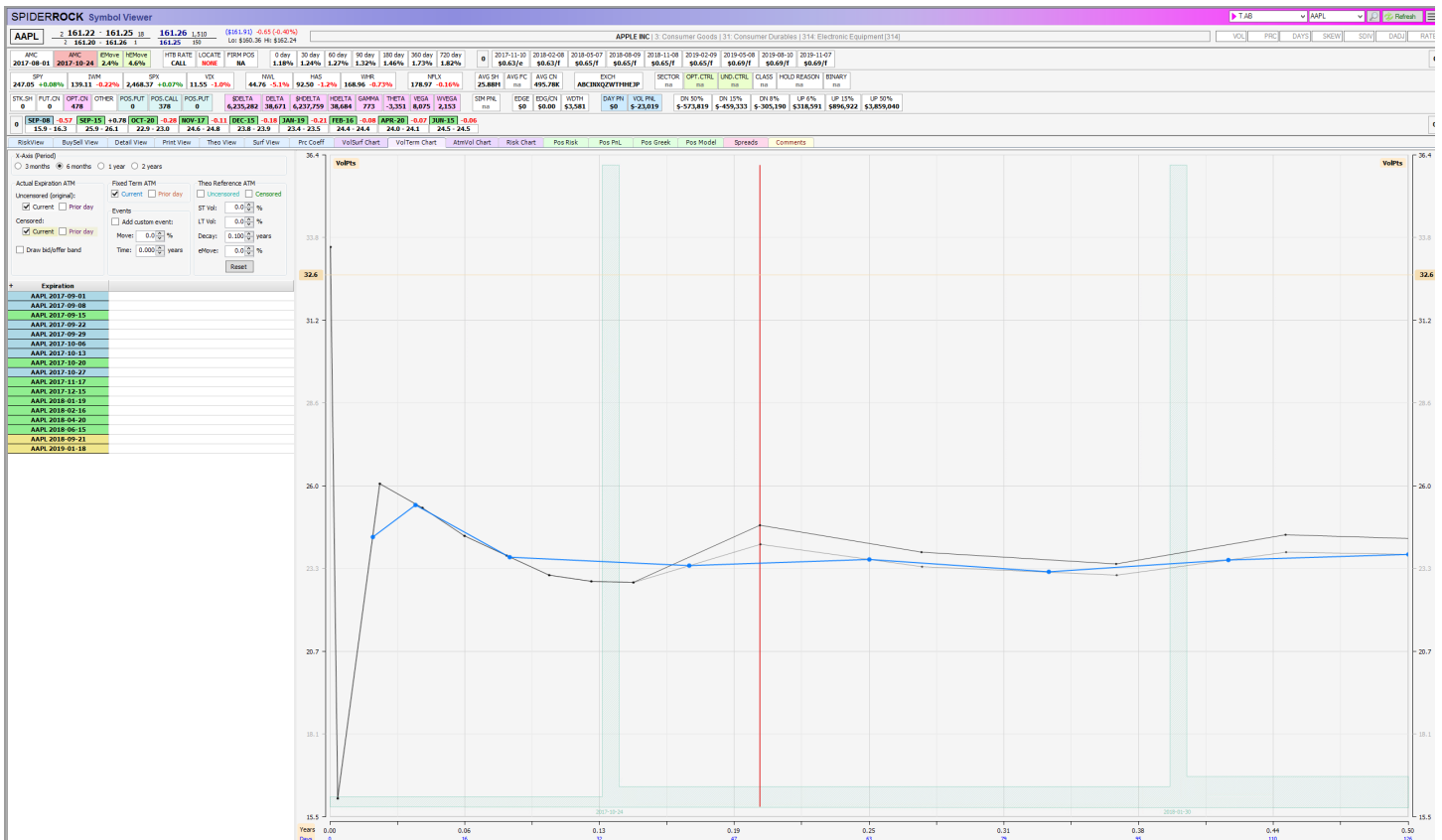
VOLSURF CHART

- The VolSurf Chart tab allows users to visualize the SpiderRock calculated volatility surface, which is fit dynamically several times per seconds through the bid/ask channel for a particular symbol/expiry.
- Puts and calls are visually represented by the colored cones formed by the vertical lines. The height represents the width of the market. If the Current view setting is set to Best (best market), the user will typically see the smooth surface fall between the prevailing bid/ask spread of each strike.
- In the Charting option panel, users can select the expiration from the pull-down menu or click an expiration from the Expiration panel. Users can also select current market views (All, Best, etc), overlay the previous vol surface, overlay user defined theos vols to the SpiderRock surface, and also display a diagram of the current positions and daily volume in the particular symbol/expiry under the surface chart.
- In addition, users can select different X-Axis and Y-Axis display options: view all strikes or the ones that are at +/- 1, 2 or 5 STD from the ATM strike. View outright vol, vol skew (centered on 0), vol difference, or price difference (vol difference * vega).
- For additional metrics related to the calculation of the surface, users can click on the 'More' button (see below)



VOLTERM CHART TAB

- The VOLTERM chart tab allows users to see a visual representation of actual term structure (uncensored and/or censored for earnings) as well as a fixed term structure of the ATM volatility for a specific symbol across a specified length of time.
- The X-Axis displays the length of time in days and years. The time display choices are 3 months, 6 months, 1 year, or 2 years.
- The Y-Axis represents ATM volatility in volatility points.
- Users can add a custom event to the chart and see how such event would impact the term structure by choosing the magnitude of the move and the time frame, in years.
- User can also overlay a "big picture" curve by adjusting the short term vol (ST Vol), long term vol (LT Vol), decay (Decay) and earning move (eMove). This "big picture" or theoretical curve can be censored or uncensored.



ATMVOL CHART TAB

- The ATMVOL chart tab allows users to have a 5-minute snapshot visual representation of the SpiderRock ATM volatility surface for up to 10 trading days.
- The base line (shown in black) is a line drawn showing the majority of time the ATM Vol spent at any one value. The vertical "T bar" line show the highest and lowest values during that time period.
- The X-Axis represents the time of day (in EST). The Y-Axis represents the values denoted below.

ivXX:	Implied volatility of fixed strike, based on prior day settlement
uPrc:	Underlying price
sDiv:	Continuous dividend rate
Slope:	Volatility surface slope
uOff:	Underlying offset (for Futures)
vsFV:	This is the variance sweep for forward volatility
log(iVol/SPY):	Implied volatility of the selected symbol over SPY implied volatility in log terms
iVol - SPY:	Implied volatility of the selected symbol minus SPY implied volatility
log(ivXX/SPY):	Implied Volatility of fixed strike over SPY implied volatility in log terms
ivXX - SPY:	Implied Volatility of fixed strike minus SPY implied volatility
log(uPrc/SPY):	Underlying price of selected symbol over SPY price in log terms

- The underlying price (last sale) and slope are captured during the same period of time. Slope is calculated as a rate of change between the volatility levels at the +/- ½ moneyness strikes.
- We also allow the ability to override the default scaling by hitting the 'Override Ranges' button

